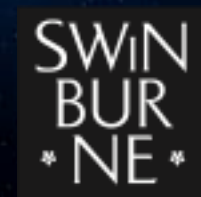


Discovering the Origins of Gravitational-Wave Sources (Heavy Double Neutron Star edition)

Confirmation of Candidature

Ashwathi Nair

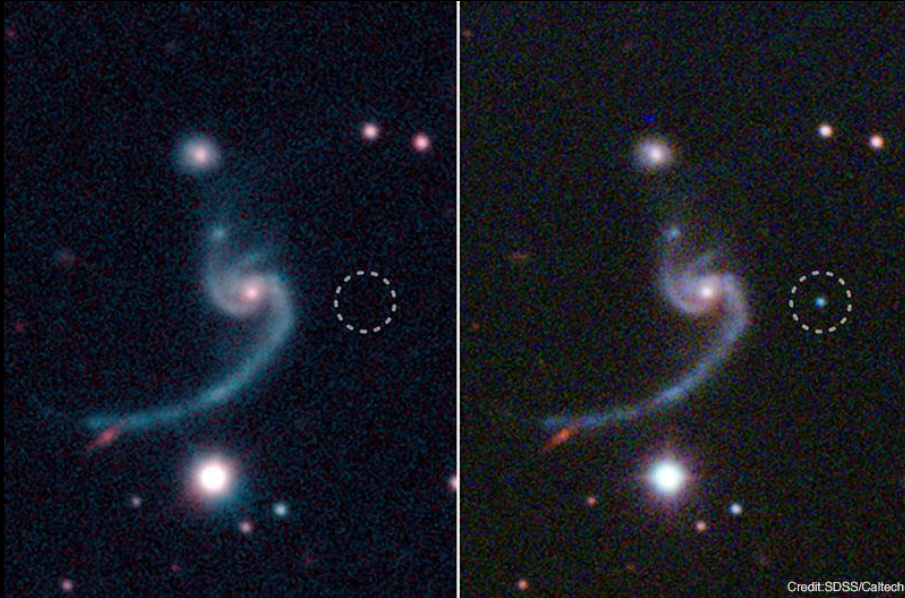
Primary Supervisor : Dr. Simon Stevenson
Secondary Supervisor : Prof. Jarrod Hurley



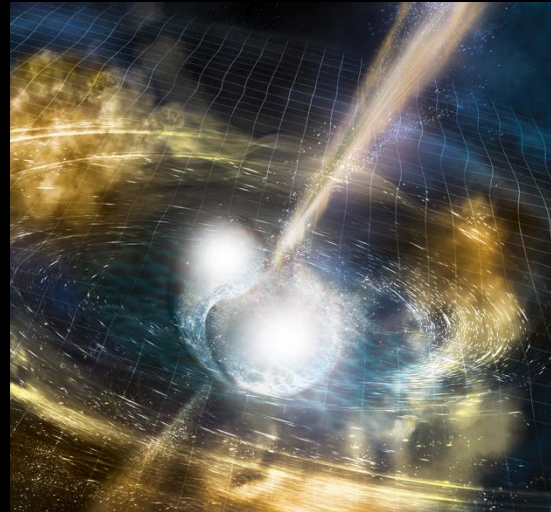
Centre for
Astrophysics and
Supercomputing

- **Introduction**
 - Motivation
 - Observational wisdom
- **Work Done (July 23 – June 24)**
 - Framework designed
- **Future work and projects**
 - Project 1
 - Project 2
 - Project 3
- **Future Timeline**

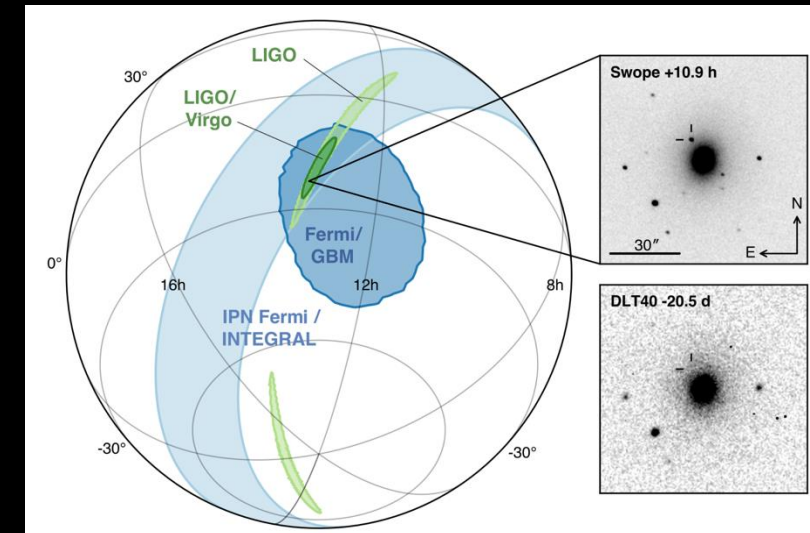
Why care about double neutron stars?



Left: Red and green composite image from the Sloan Digital Sky Survey (SDSS) taken before SN iPTF14gqr. Right: Red/green/blue composite image from Palomar 16-inch telescope taken on Oct 19, 2014. During SN iPTF14gqr. (Credit: SDSS/Caltech)



Gamma ray burst and Kilonova AT2017gfo in the galaxy NGC 4993 2secs after GW170817



What do we know about DNS systems so far ?

Detected by radio telescopes and gravitational-wave detectors

Radio Astronomy

~ 30 Galactic DNS between total mass of $2.50\text{--}2.89M_{\odot}$

Gravitational Waves

GW170817

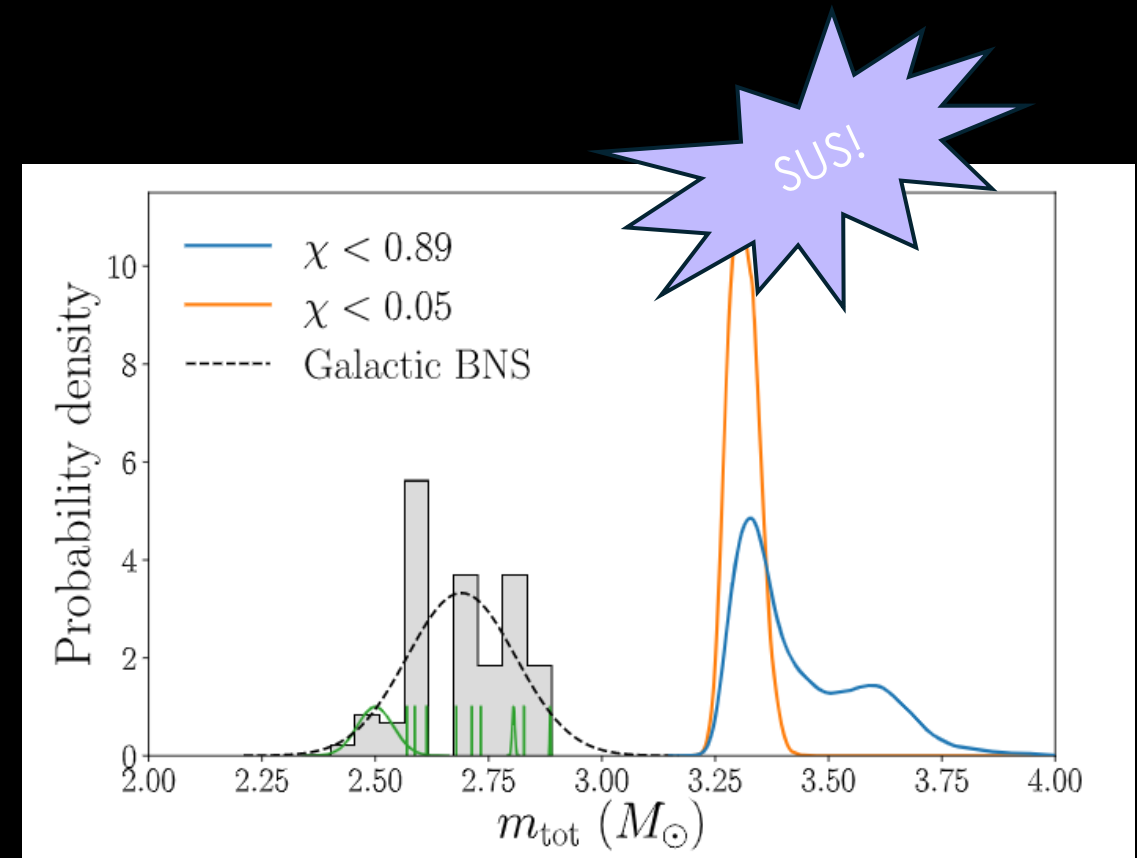
component masses : 1.17 to $1.60 M_{\odot}$

Total mass : $2.87^{+0.47}_{-0.09} M_{\odot}$

GW190425

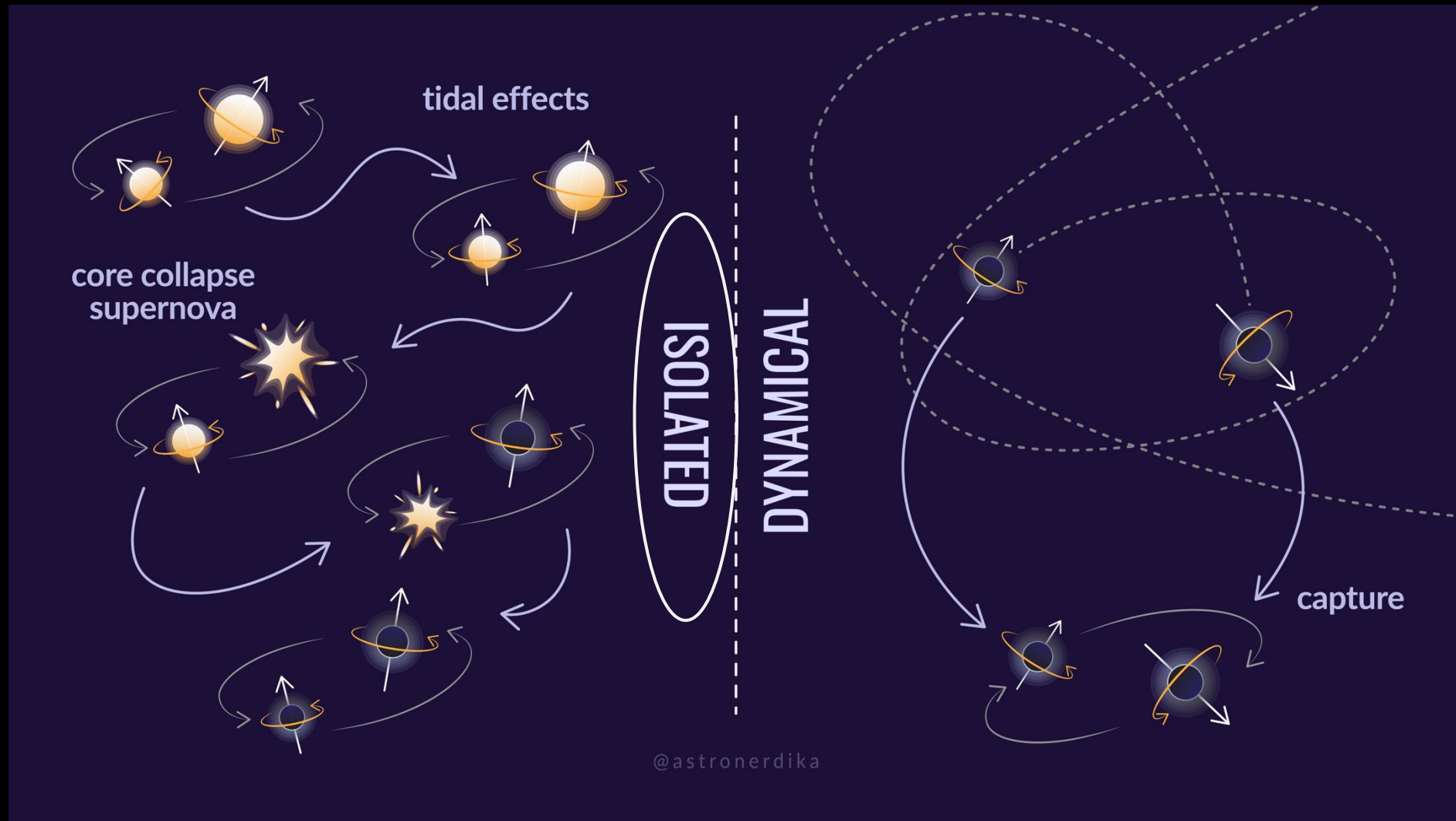
component masses : 1.1 to $2.5 M_{\odot}$

Total mass : $3.4^{+0.3}_{-0.1} M_{\odot}$

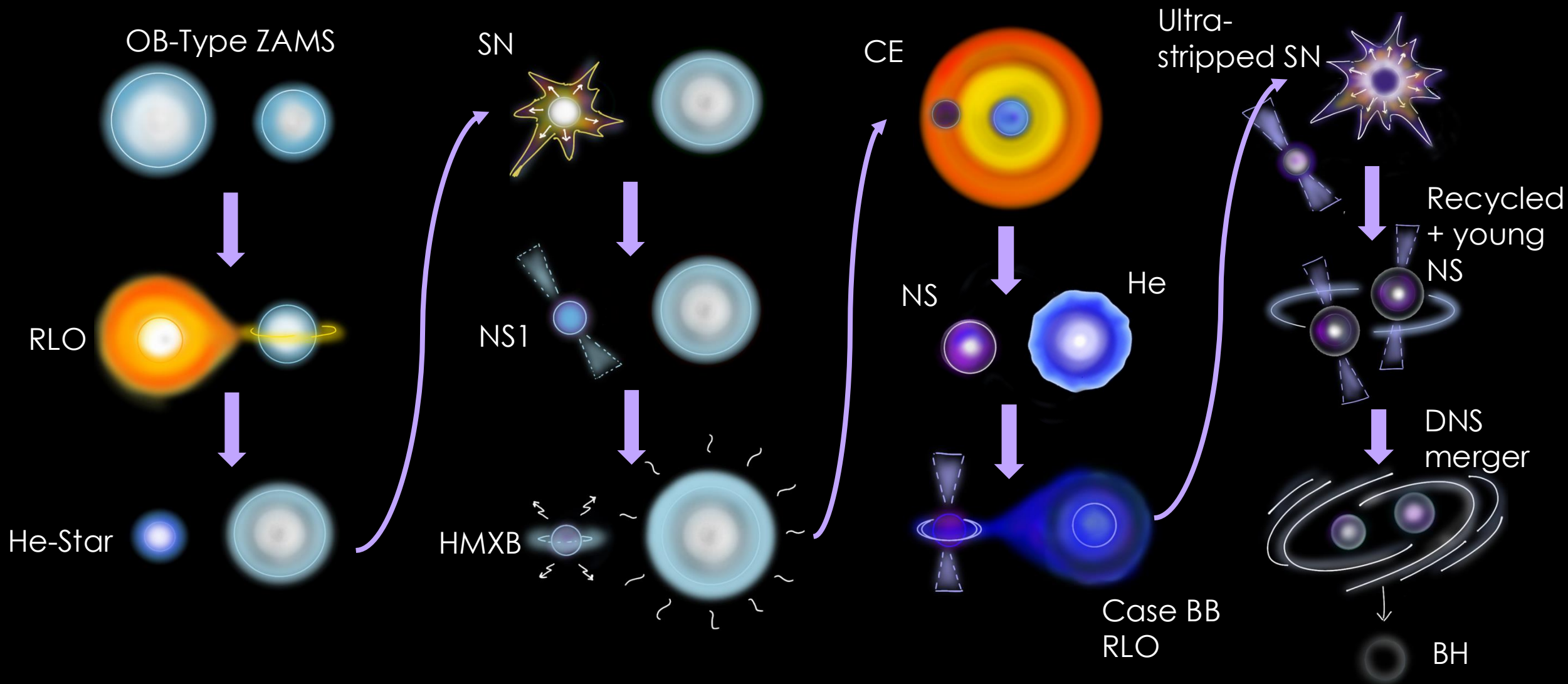


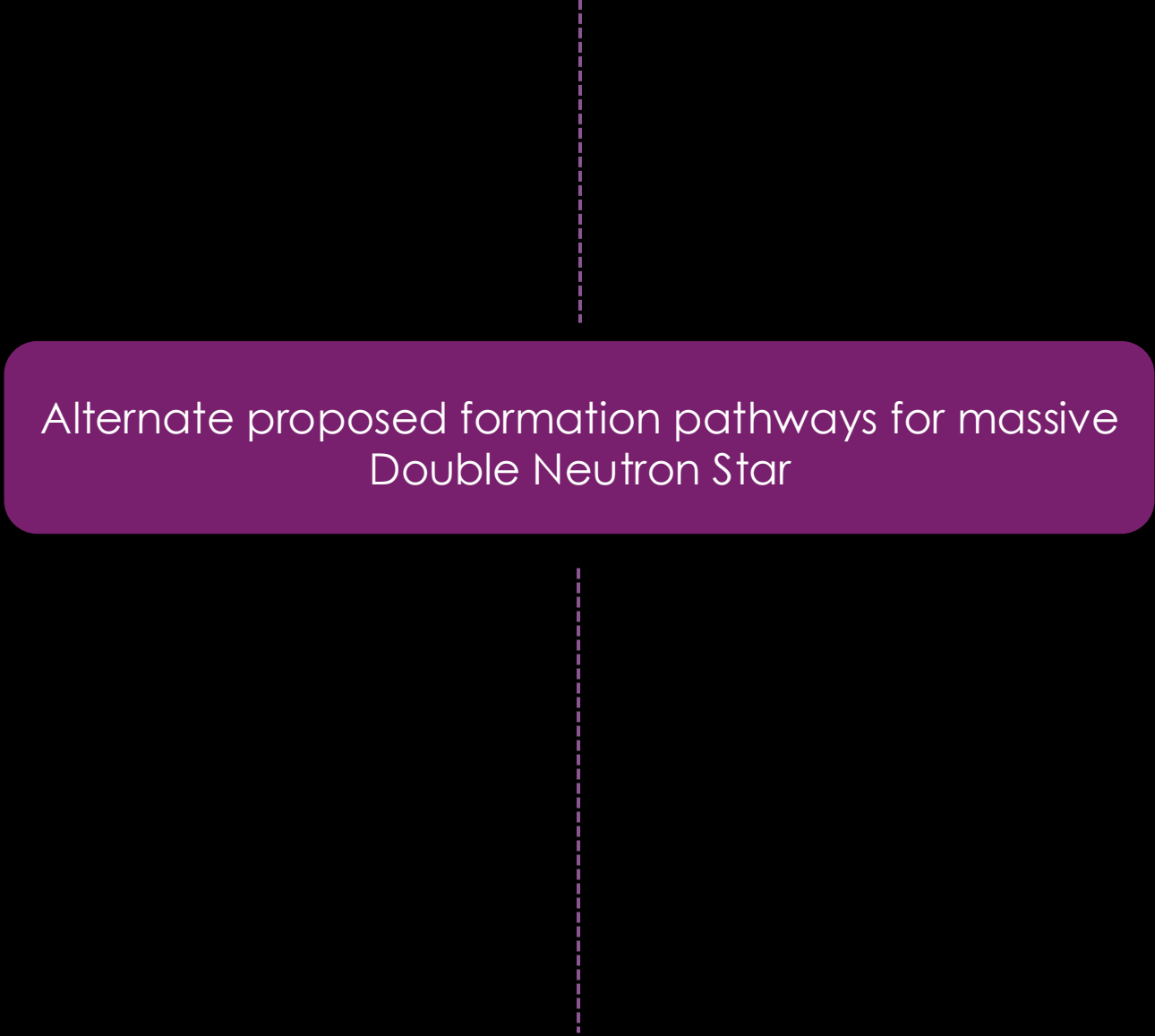
Abbot et al. 2020

Formation Channels of compact binary mergers



Standard Double Neutron Star formation pathway – T. Tauris et al. 2017



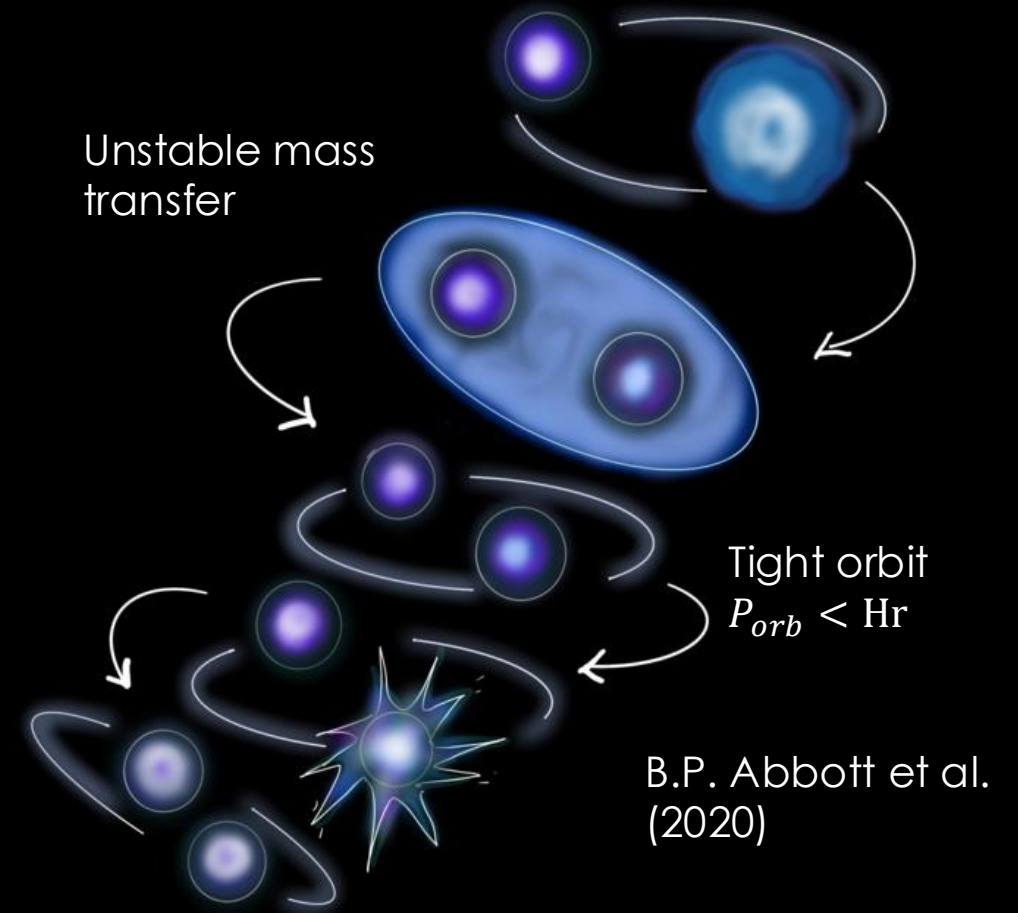


Alternate proposed formation pathways for massive
Double Neutron Star

Alternate proposed formation pathways for massive
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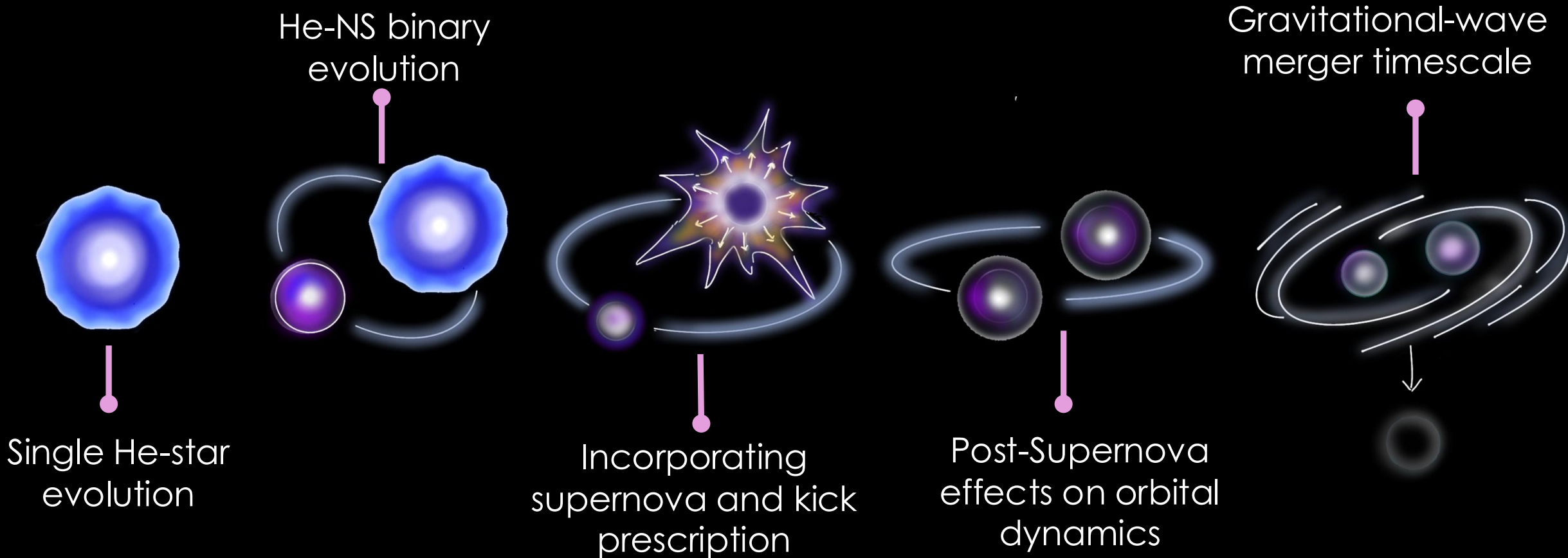


Alternate proposed formation pathways for massive Double Neutron Star

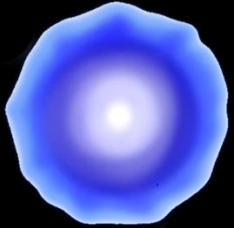


Framework created for the formation pathway

Framework created for the formation pathway



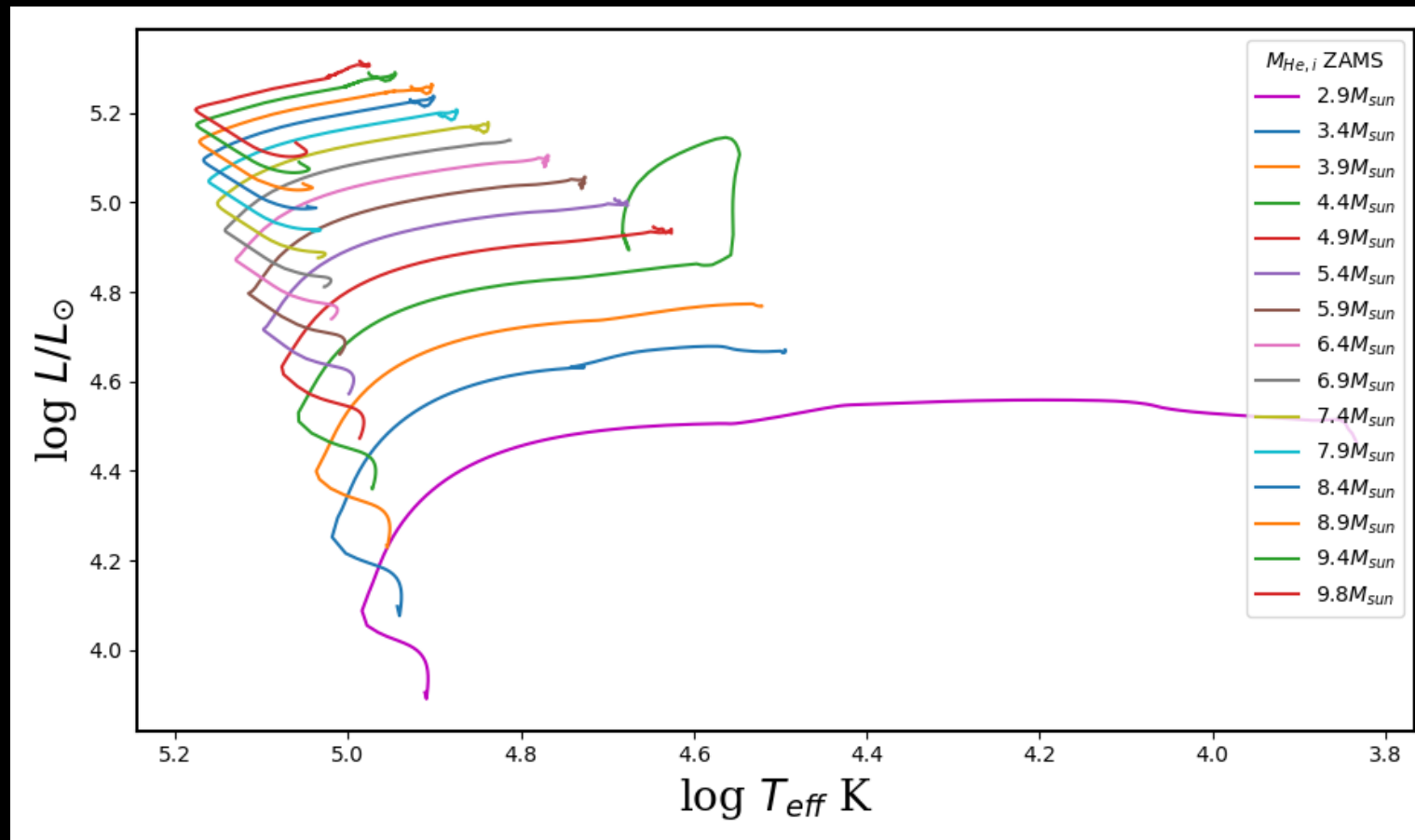
Single Helium Star Evolution



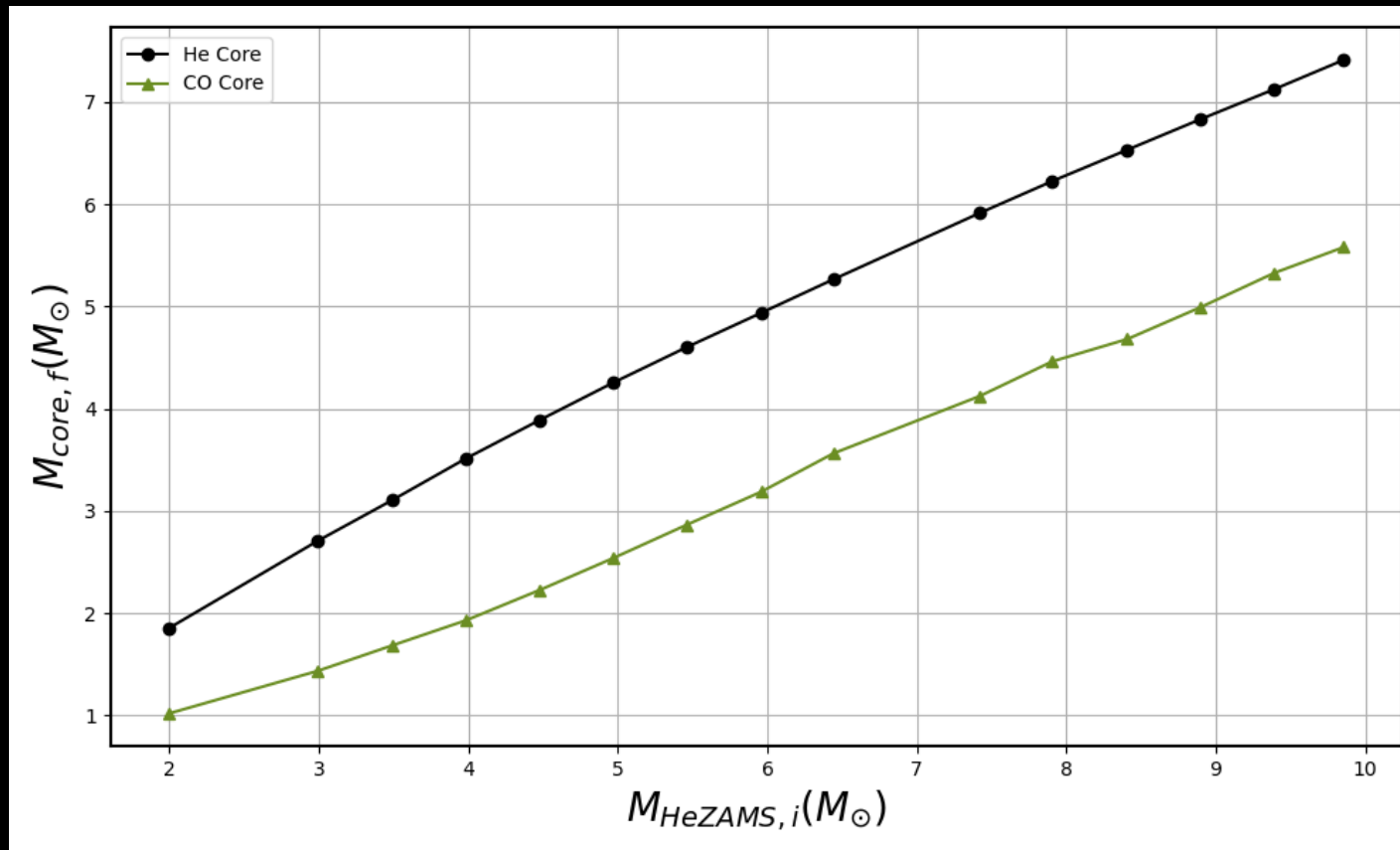
- **Simulations were performed using Module for Experiments in Stellar Astrophysics (MESA) module MESASTAR for intermediate and massive He-star ($2.9\text{--}9.8 M_{\odot}$)**
- Helium mass fraction $Y = 0.98$, at solar metallicity $Z = 0.02$
- 'Dutch' prescription for mass loss by hot winds (Nugis & Lamers 2000; Vink et al. 2001; Glebbeek et al. 2009)- scaling factor of 0.8 for massive helium stars (Meynet & Maeder 2000).
- Ledoux criterion for convection, with a mixing length alpha of factor 2, and alpha semi-convection of 1.
- Gold tolerances for intermediate and massive Helium star models, Type2 opacities for $Z = 0.02$



Hertzprung-Russell (HR) diagram of single helium stars with masses ranging from $2.9M_{\odot}$ - $9.8M_{\odot}$

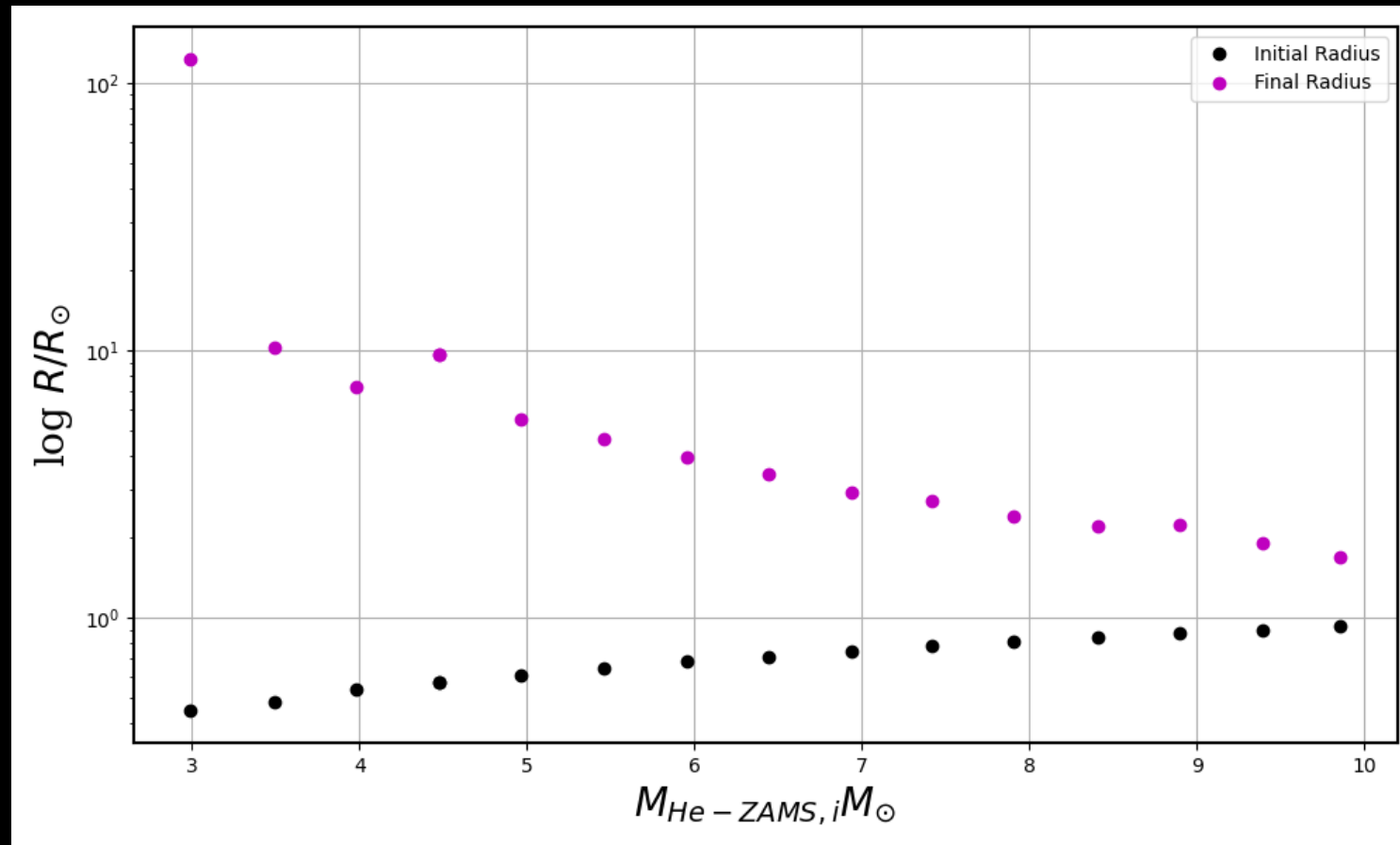


He and carbon-oxygen (CO) core masses of all He-stars

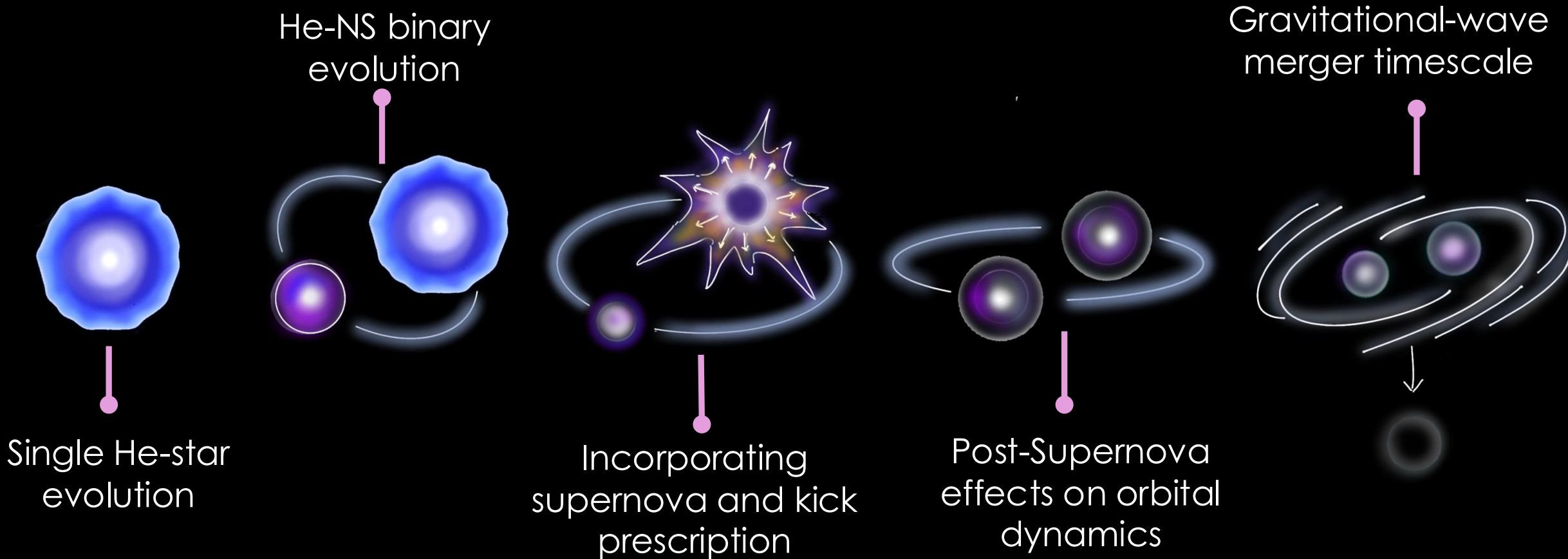


- Core masses are well determined by the star's initial mass and standard choice of stellar physics.

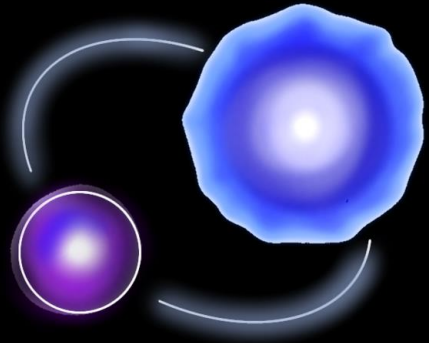
Initial and final radius as a function of initial He-ZAMS mass



Framework created for the formation pathway

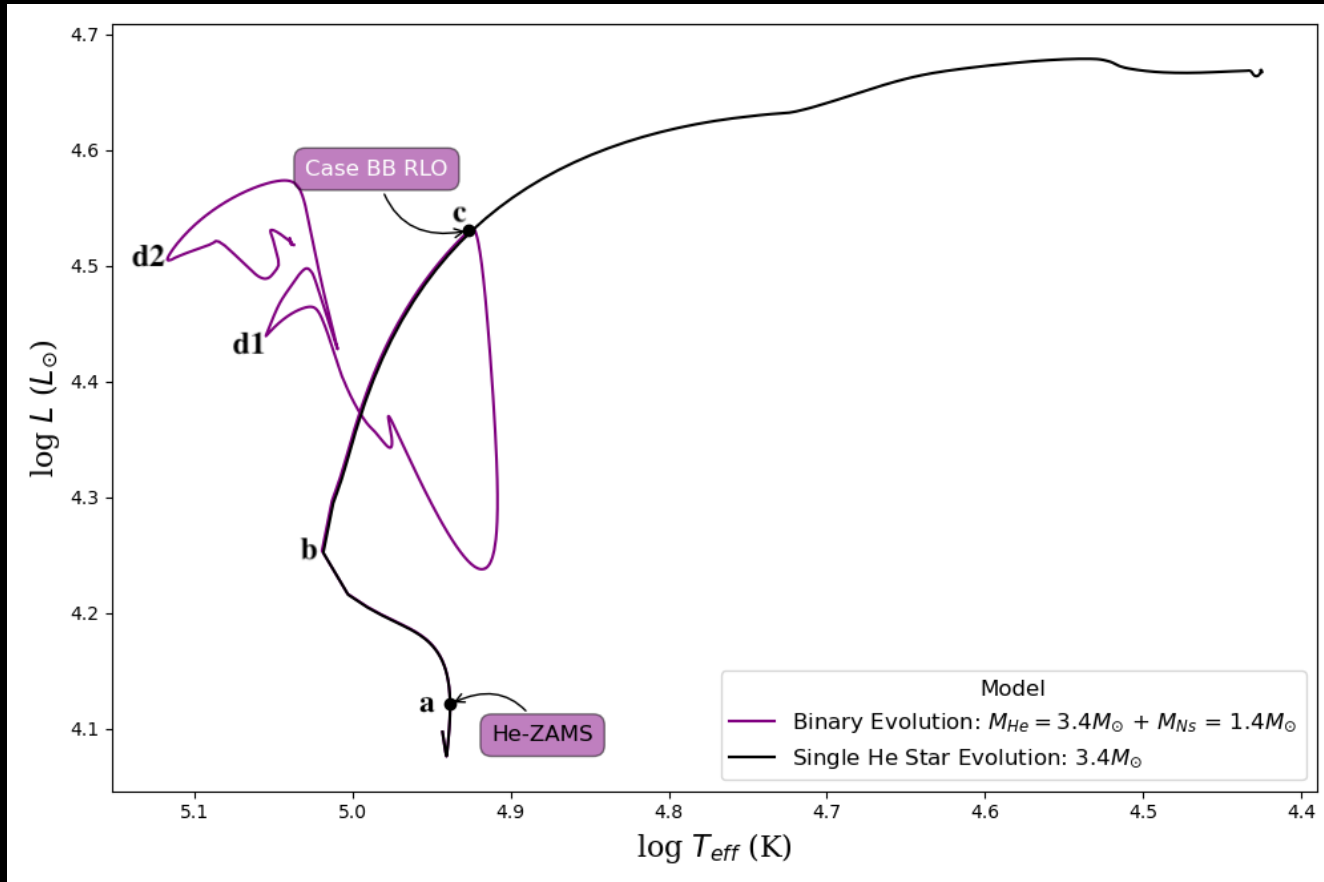


Binary Star Evolution

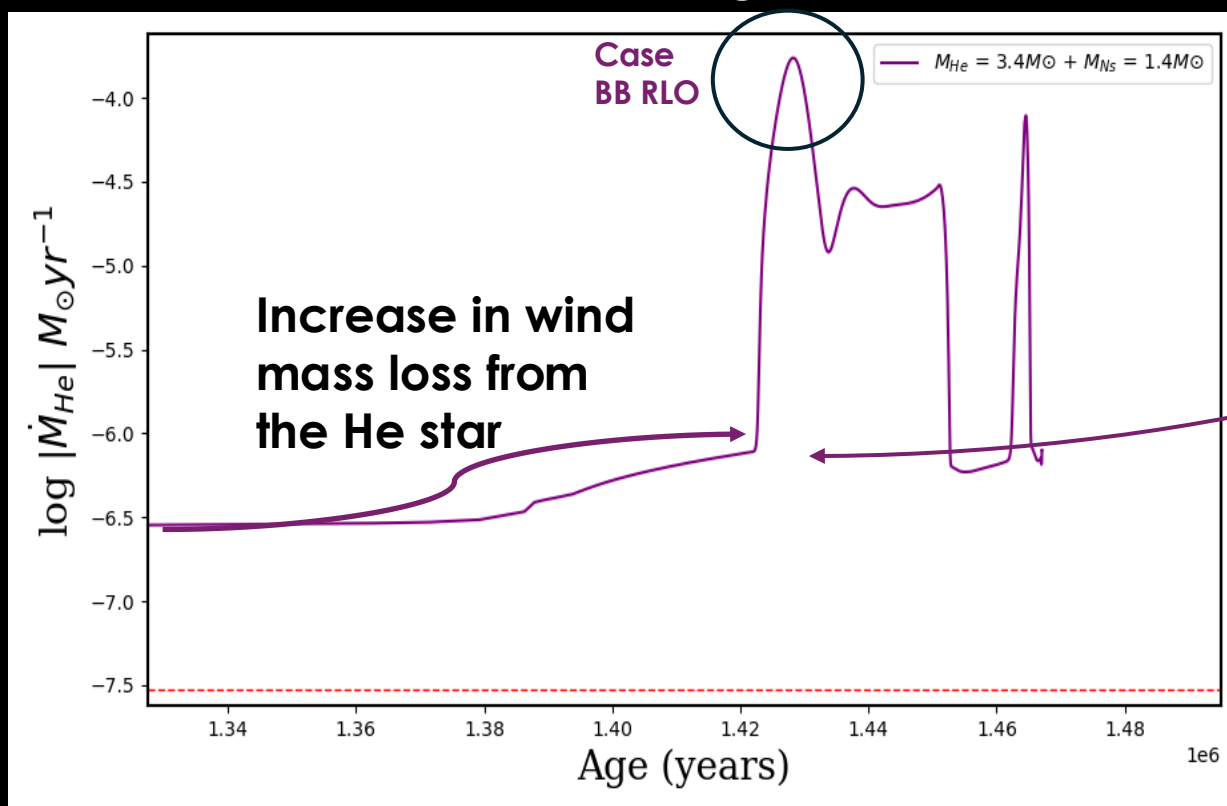


- Modelled the binary evolution of a ZAMS He-star with a $1.4M_{\odot}$ NS companion using MESAbinary
- The binary is initially set up in a circular orbit, with an initial orbital period of 3 hours.
- 'Ritter' mass transfer scheme (Ritter 1988) is used to calculate the mass transfer rate from the He star.

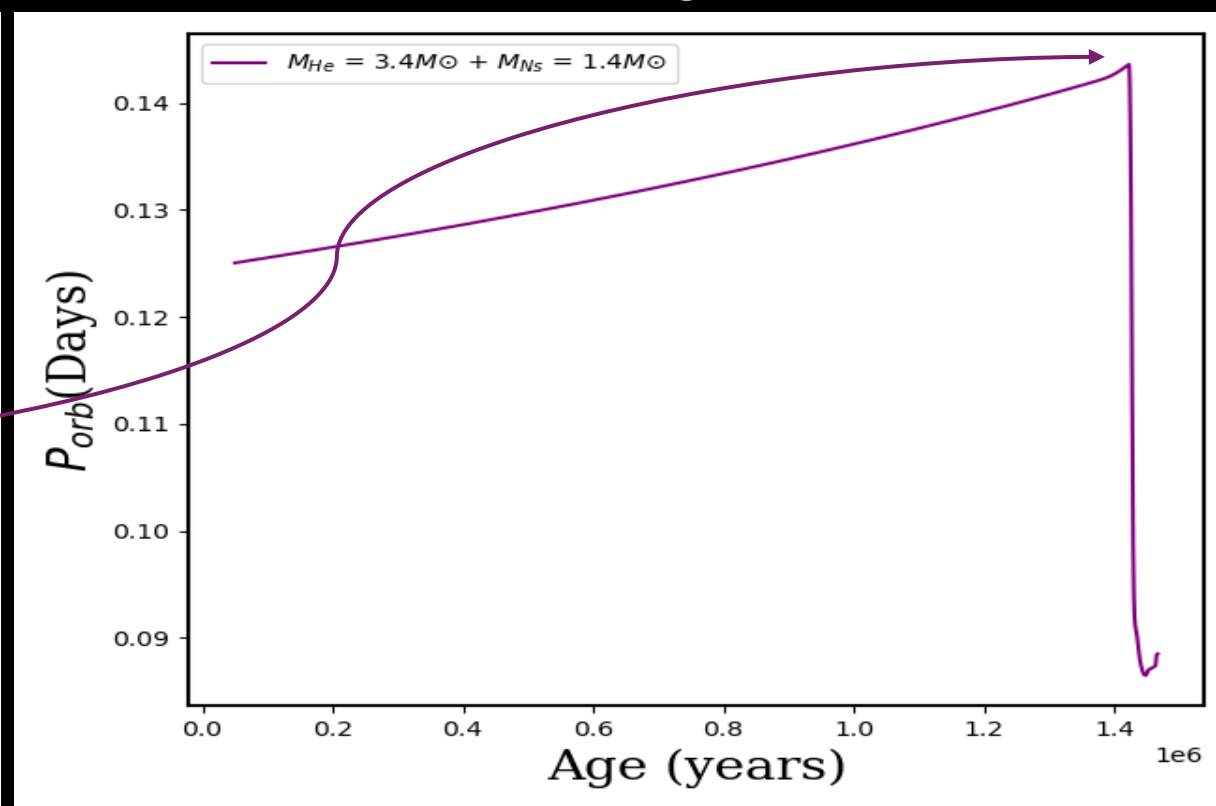
HRD of $3.4 M_{\odot}$ He + $1.4 M_{\odot}$ NS binary



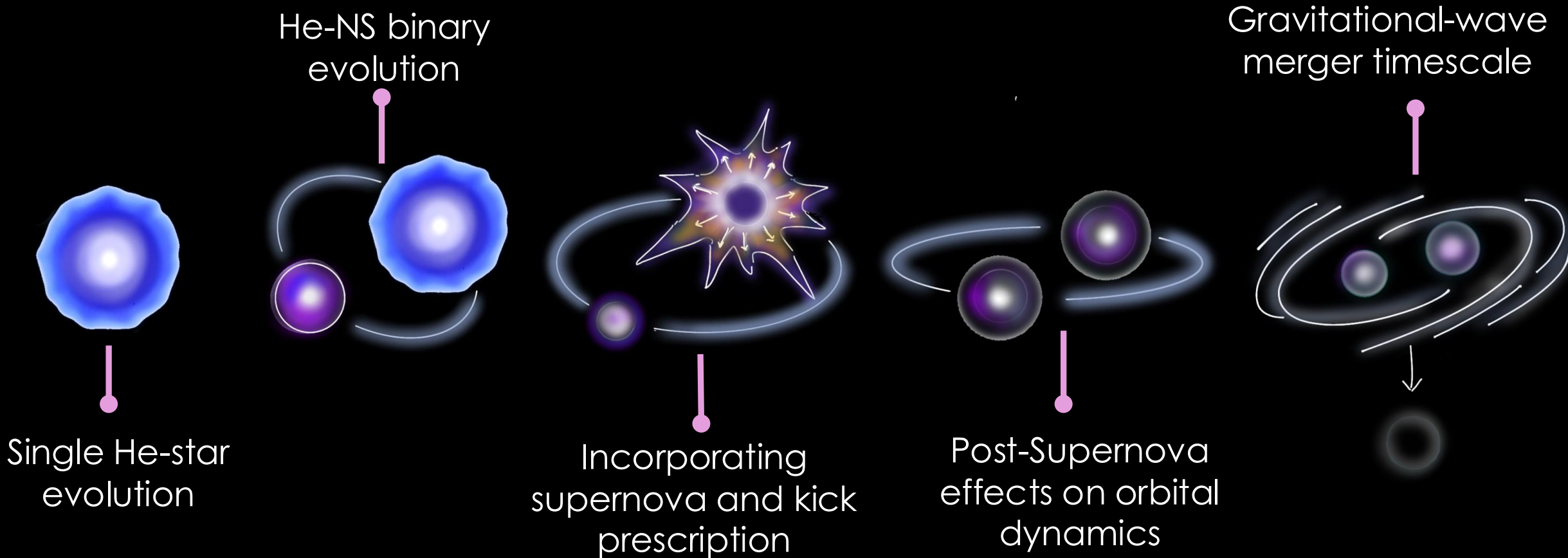
Wind mass loss from the He star as a function of the star's age



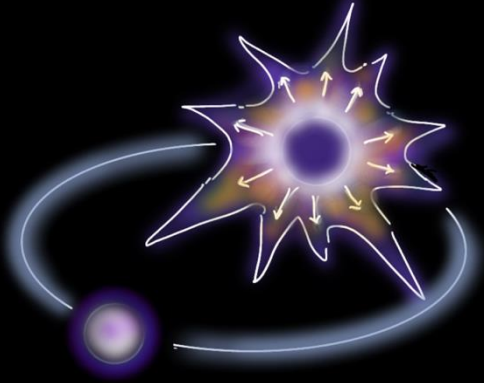
Change in orbital period as a function of the star's age



Framework created for the formation pathway

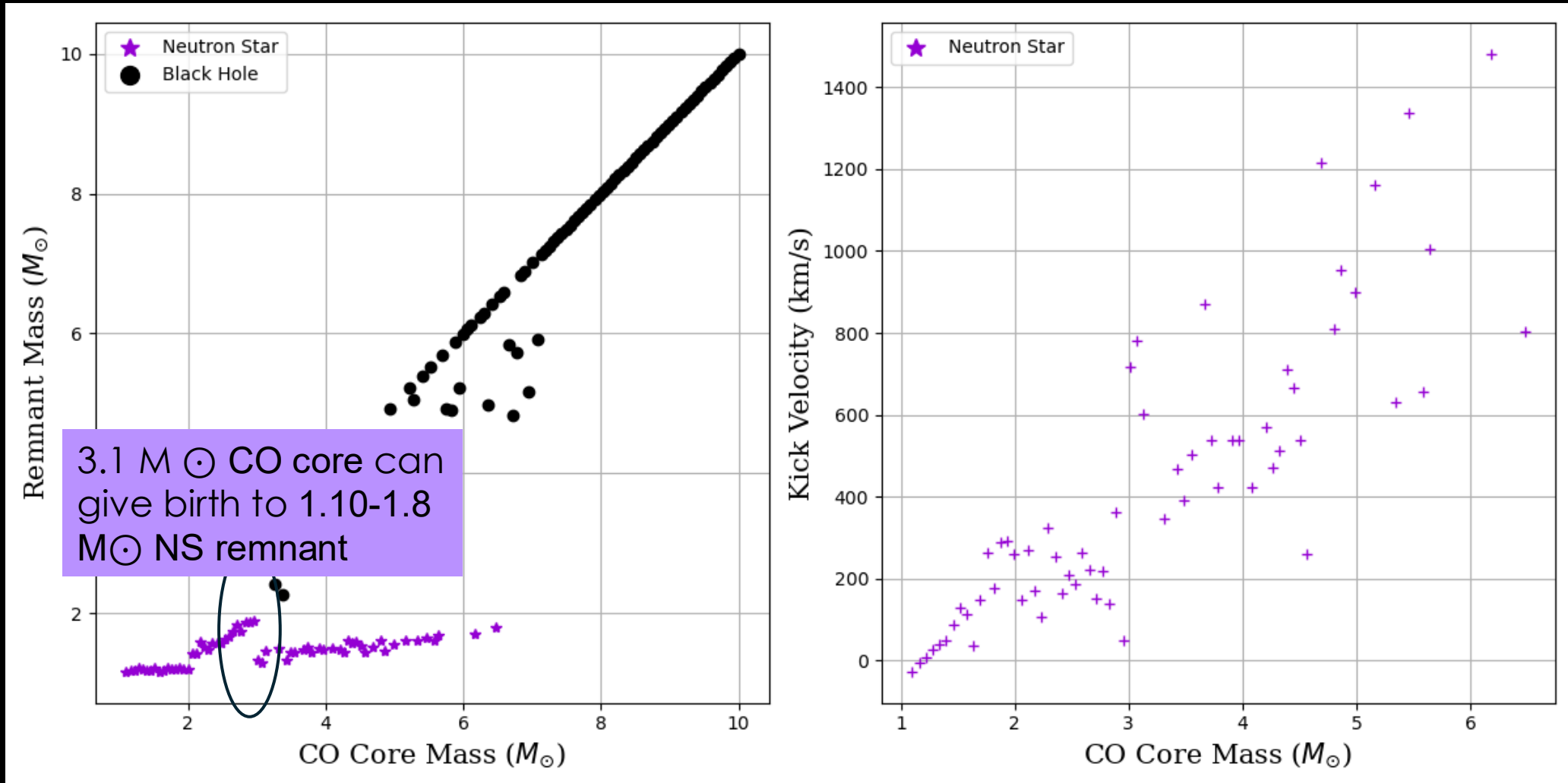


Incorporating supernova and kick prescription

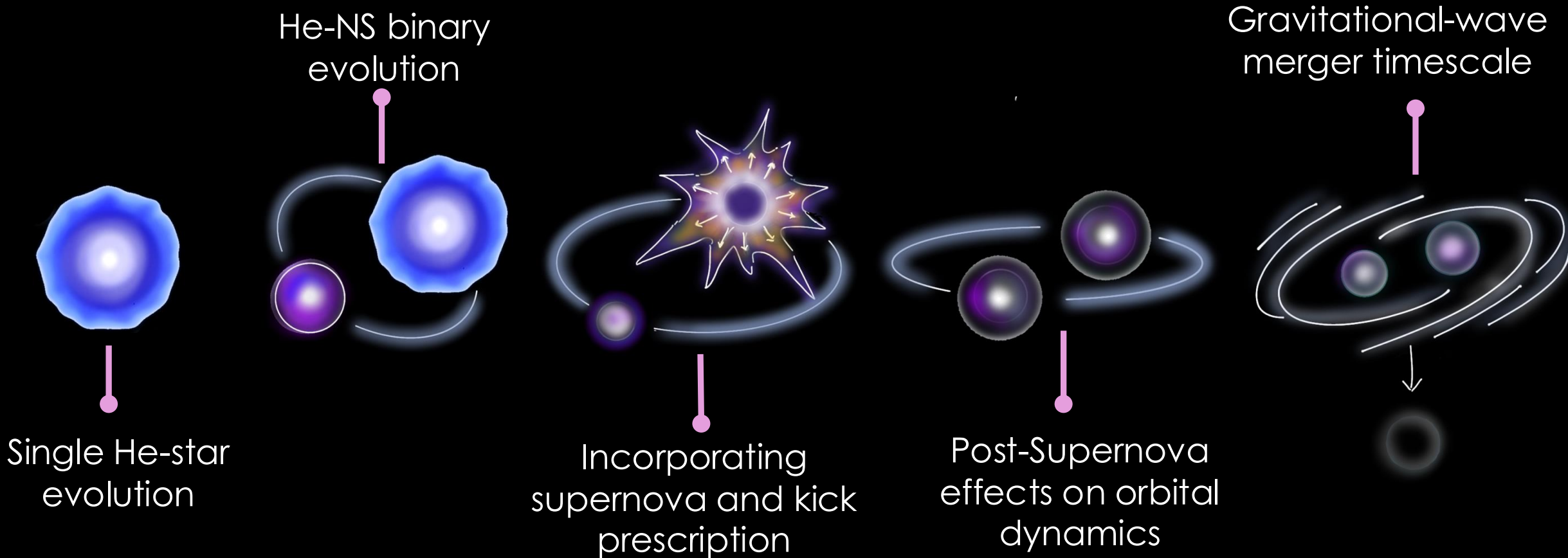


- Mandel & Muller (2020) have developed an analytic prescription for the dependence of the mass of NS and BHs and natal kicks based on the pre-SN carbon-oxygen (CO) core and helium shell mass

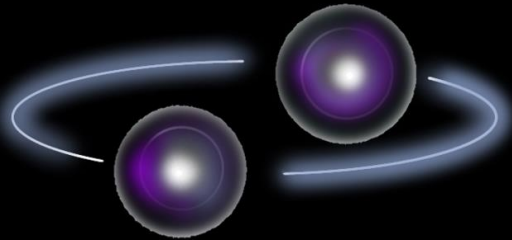
Remnant masses and Kick velocity as a function of CO core masses



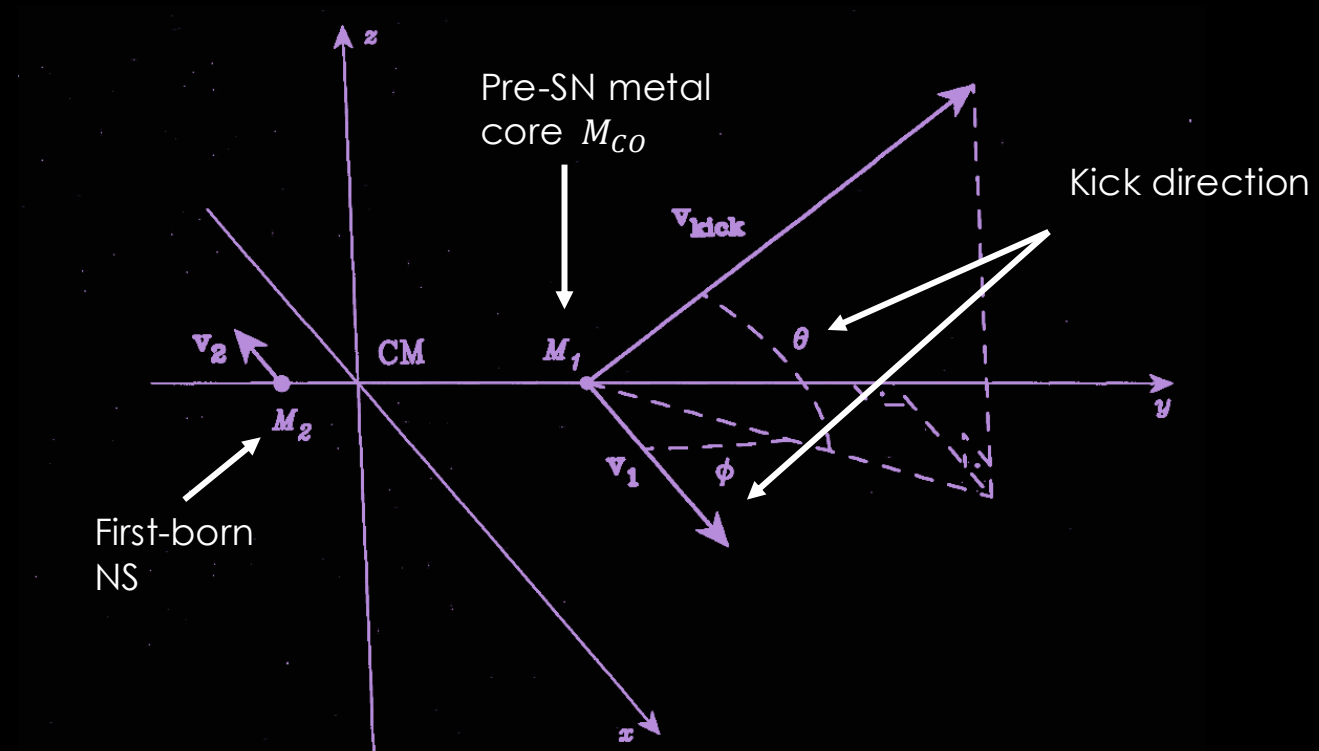
Framework created for the formation pathway



Post-Supernova effects on orbital dynamics

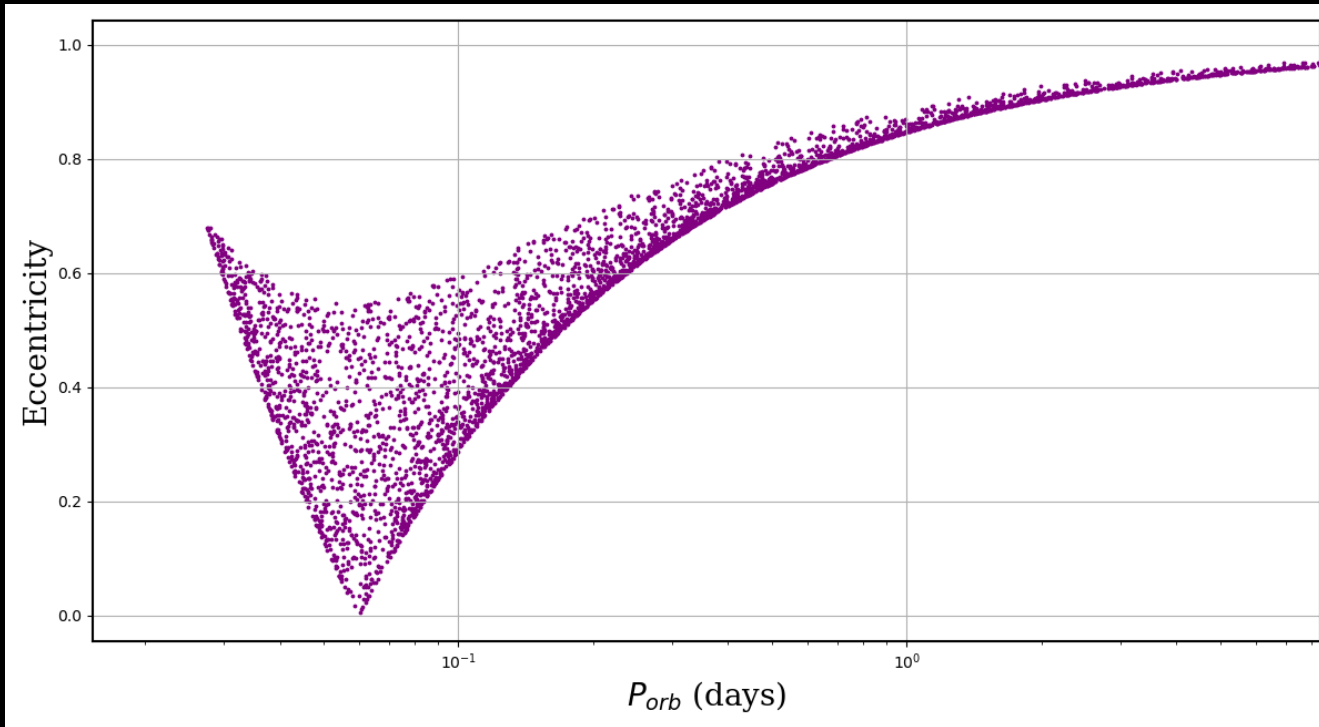


Determine the relation between the initial and final binary orbital parameters and the effect of asymmetric supernovae on binary systems



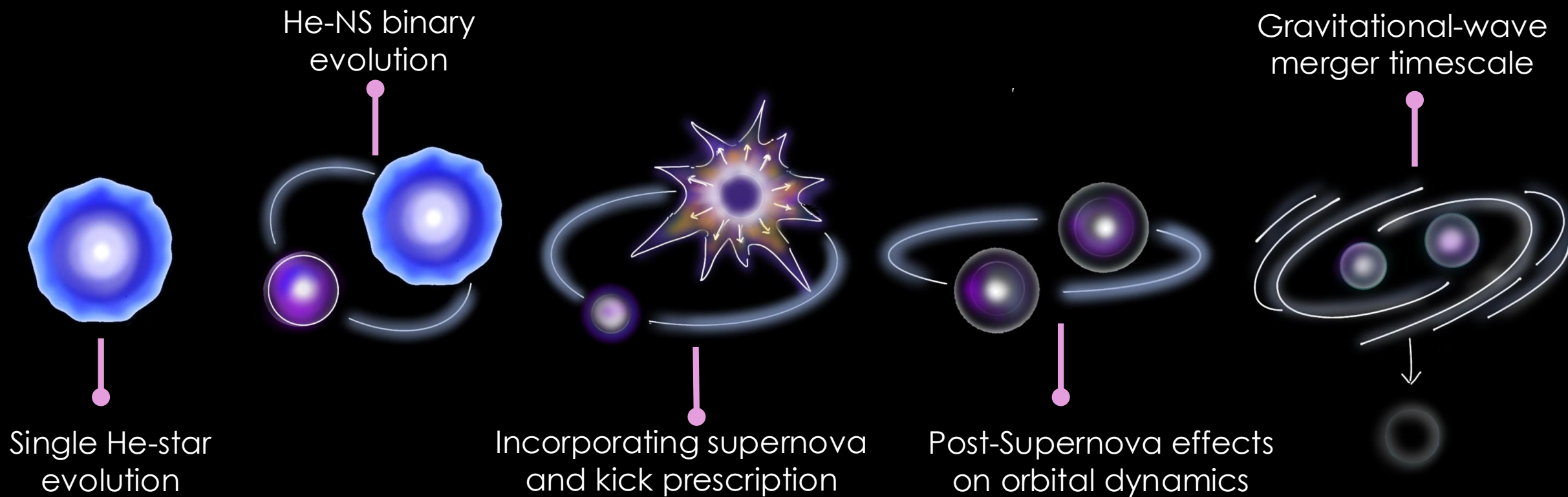
Brandt et al. 1995

Eccentricity as a function of orbital period for Model 3: $M' = 1.5M_{\odot} + 1.4 M_{\odot} \text{ NS}$ (post-SN binary)

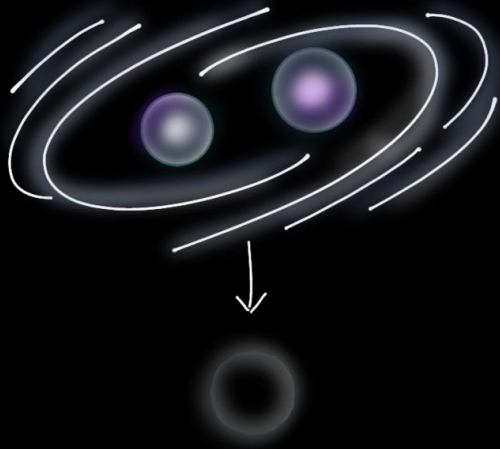


The *post-SN* semi-major axis, orbital period, and the *post-SN* eccentricity determine if the binary will merge within the Hubble time.

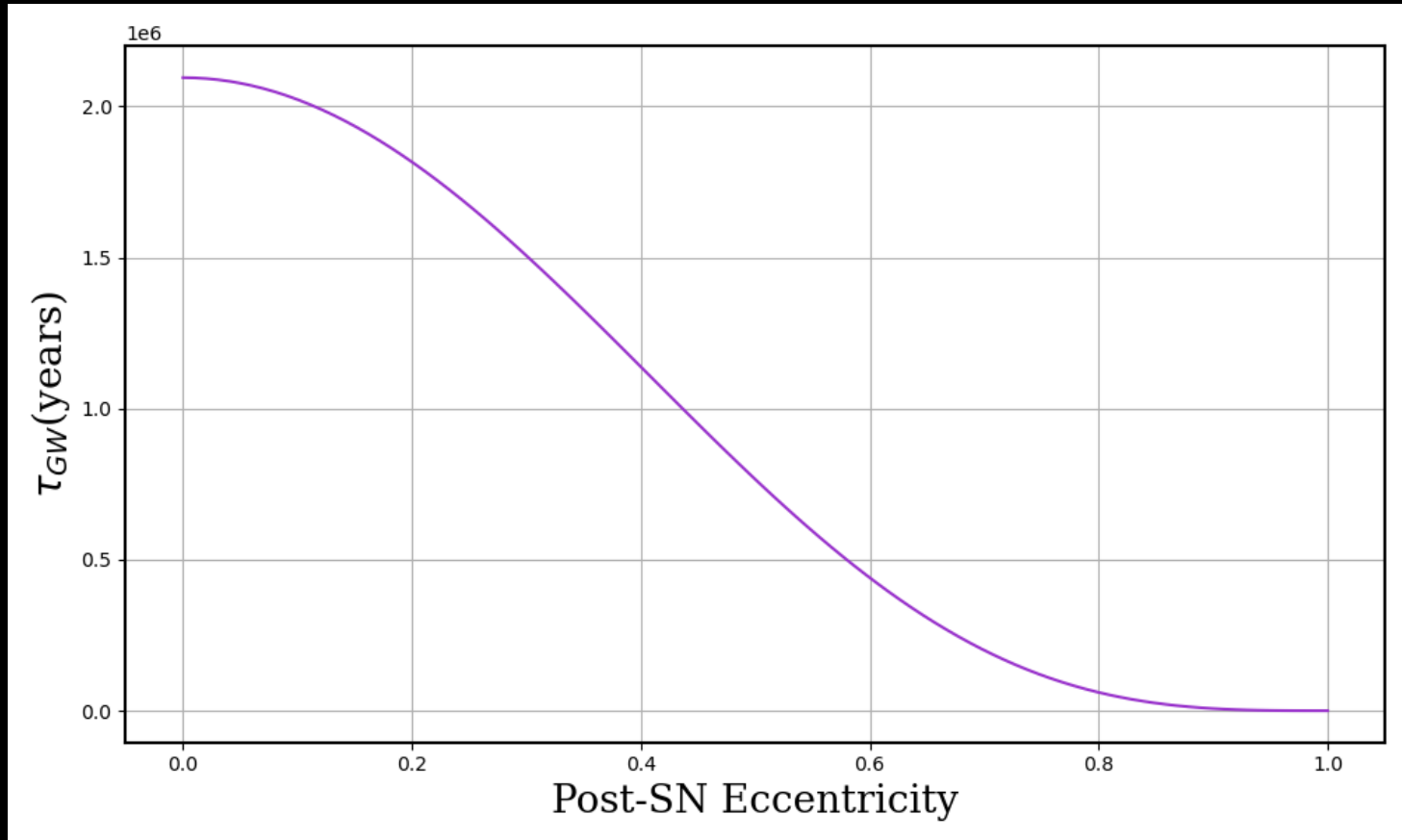
Framework created for the formation pathway



Gravitational-wave merger timescale



Gravitational wave merger time as a function of post-SN eccentricity for $\text{NS2} = 1.5 M_{\odot} + \text{NS1} = 1.4 M_{\odot}$



Project 1 and 2

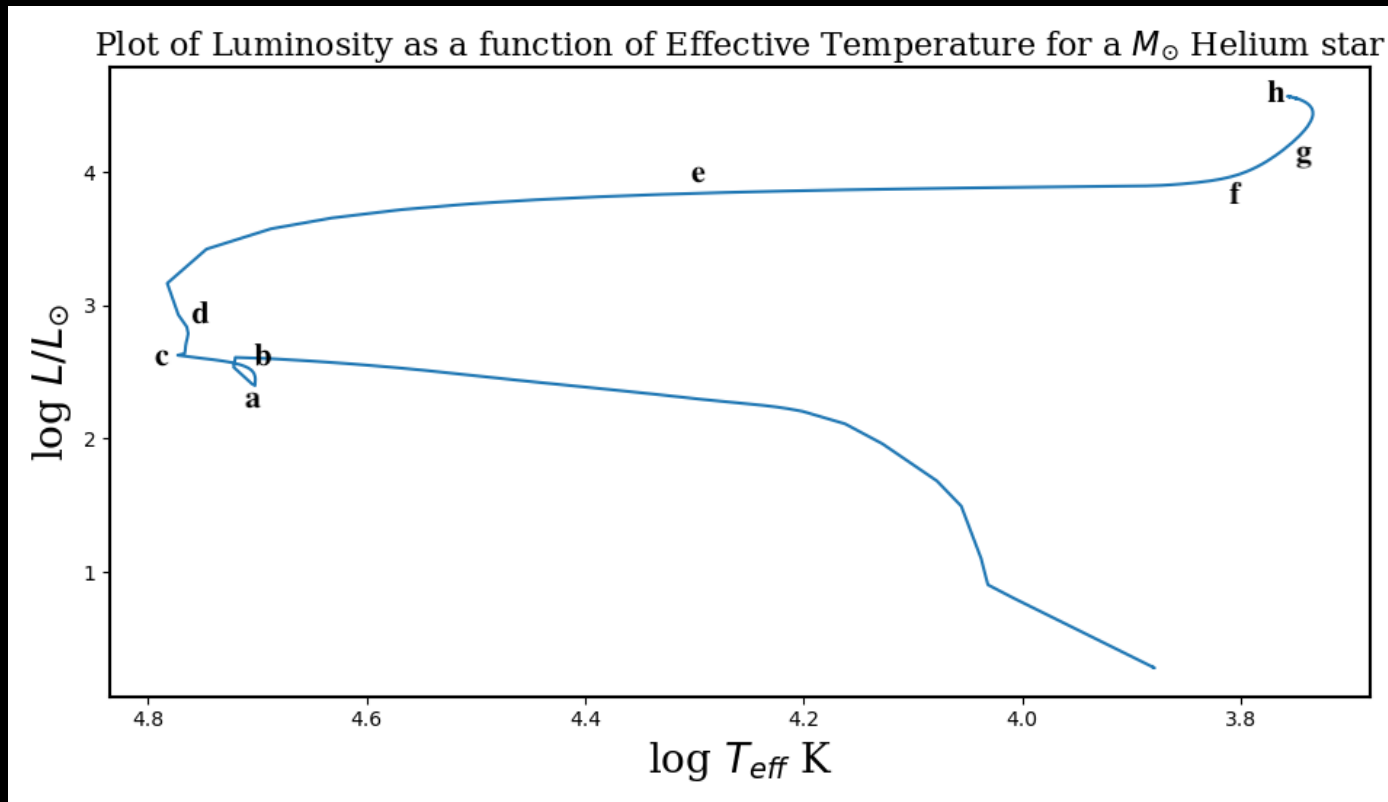
- Run a grid of MESA binary models for the initial parameter space, over masses and orbital periods to probe the properties of the binary
- Use the framework to model the supernova and calculate the gravitational-wave merger timescale.
- Population synthesis using COMPAS
- Compute the formation rates and mass distributions of heavy double neutron stars from various formation channels and do a comparative study, to help determine the most likely formation channel for heavy double neutron stars.
- Compile an up-to-date list of the observed Galactic DNSs. This will result in Paper 1.
- Paper 2 detailed follow-up of the current work. Additionally, examine further model predictions, such as the locations of mergers in host galaxies and predictions for O5.

Project 3 (Ideas)

- If the O4b run were to detect a BNS detection - examine its formation channel
- Incorporate my helium models with Method of Interpolation for Single Star Evolution (METISSE) code by Agrawal et al. 2020, and carry out population synthesis with METISSE
- Dynamical formation of double neutron stars
- A project dedicated to studying Neutron Star-White Dwarf NS-WD binaries for Laser Interferometer Space Antenna (LISA)

Year	Month Range	Activity
2024	Today	Confirmation of Candidature Review
	June	Winter Schools and Conferences
	July	Finalizing Project 1 results
	Aug - Sept	Paper 1 preparation and submission
	Oct - Dec	Project 2
2025	Jan - May	Project 2
	June	Mid-candidature Review
	May - July	Paper 2 preparation
	July - Aug	Paper 2 Submission
	Sept - Dec	Project 3
2026	Jan - June	Paper 3 preparation
	June	Draft Thesis Review
	July	Paper 3 submission
	June - Nov	Thesis preparation
	Dec	Thesis Submission

1 $M_{\odot}$ Helium Star Evolution



a : zero age main sequence

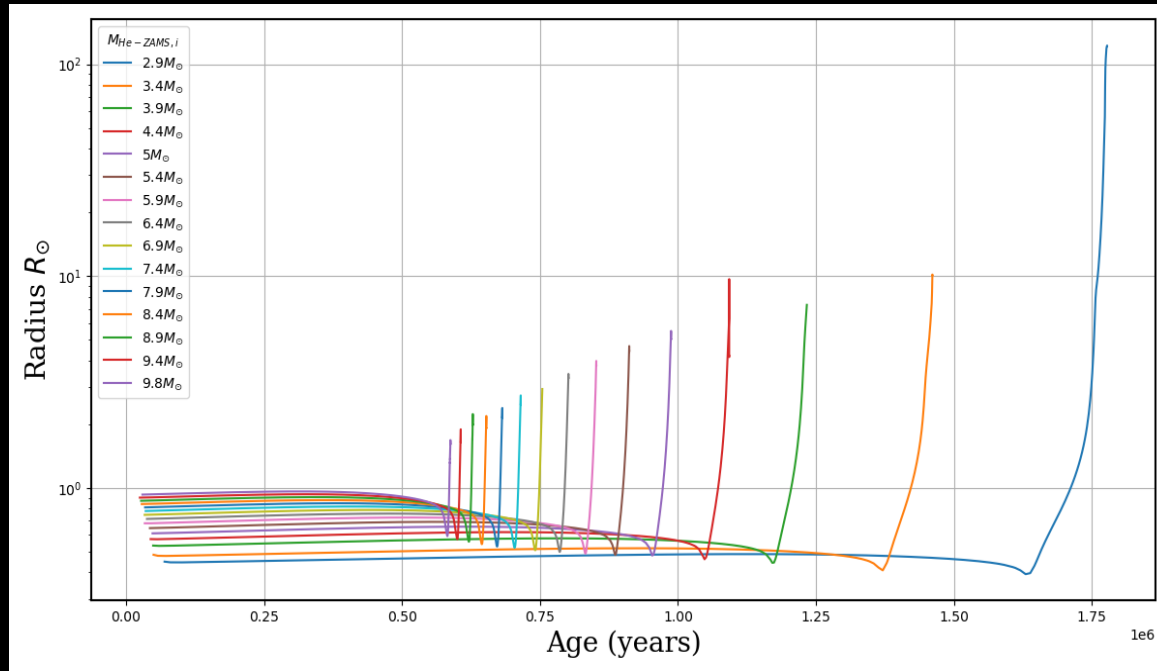
a - b : Main sequence phase

b - c : exhaustion of central helium,

c - d : first cooling phase,

d - e : slow phase of shell helium burning

e - h : fast phase of helium shell burning



Model	M_{CO} ($M_{\odot}$)	M' ($M_{\odot}$)	$P_{\text{orb,pre-SN}}$ (d)	a_i ($R_{\odot}$)	v_{kick} (km s^{-1})	Kick Direction (ϕ, θ)(rad)		a_f ($R_{\odot}$)	$P_{\text{orb,post-SN}}$ (d)	e
1	1.4	1.2	0.129	1.52	~ 80	2π	π	1.27	0.1	0.03
2	1.68	1.22	0.08	1.23	~ 174	2π	π	0.9	0.06	0.1
3	3.1	1.5	0.04	0.92	~ 532	2π	π	0.54	0.02	0.4

The *post-SN semi-major axis* a_f , *orbital period* $P_{\text{orb,post-SN}}$, and the *post-SN eccentricity* e are important to figure out if the binary will merge within the Hubble time.